Assignment No.8

Title: Set up a local Kubernetes cluster with Minikube to deploy and scale your Dockerized AI application.

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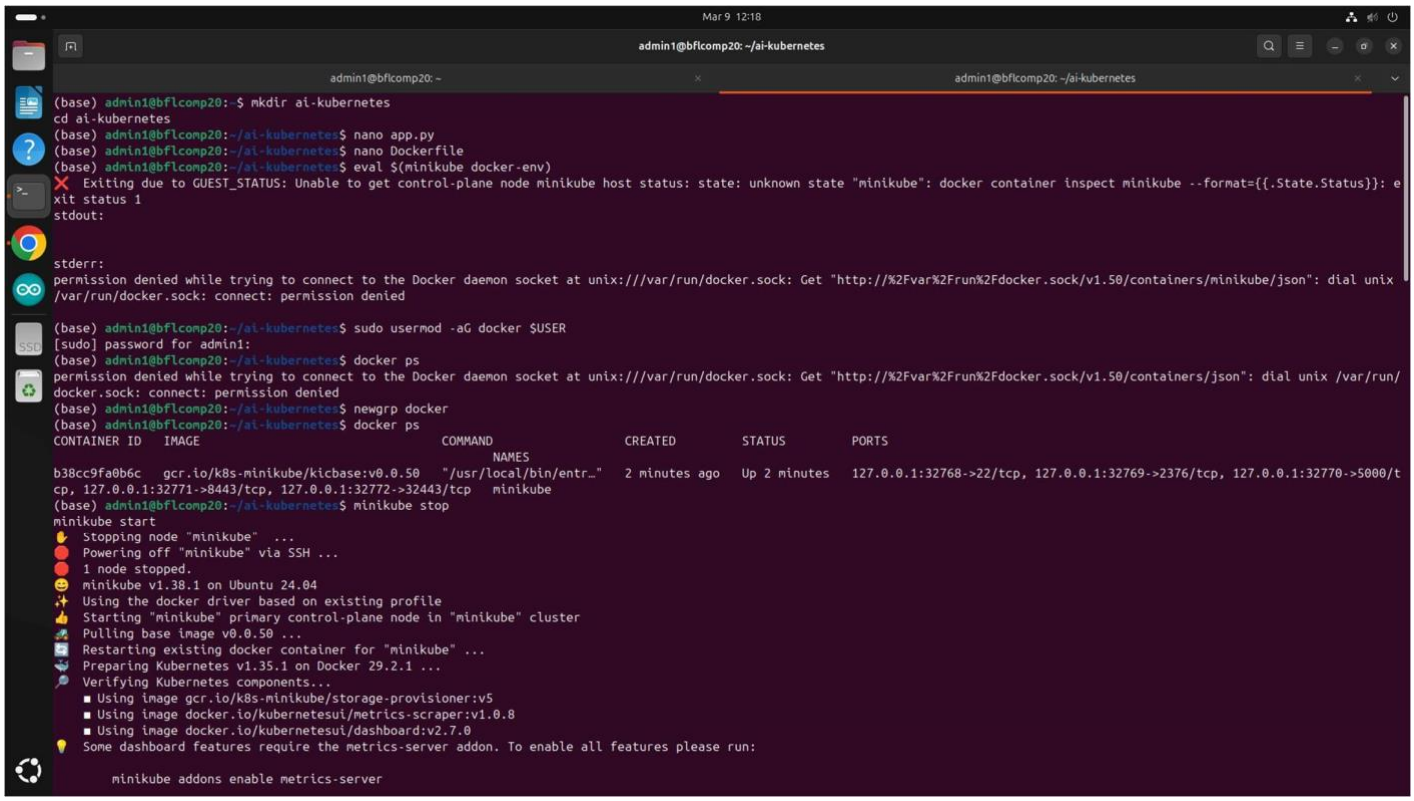
Output:



```
(base) admin1@bflcomp20:~$ mkdir ai-kubernetes
cd ai-kubernetes
(base) admin1@bflcomp20:~/ai-kubernetes$ nano app.py
(base) admin1@bflcomp20:~/ai-kubernetes$ nano Dockerfile
(base) admin1@bflcomp20:~/ai-kubernetes$ eval $(minikube docker-env)
❌  Exiting due to GUEST_STATUS: Unable to get control-plane node minikube host status: state: unknown state "minikube": docker container inspect minikube --format={{.State.Status}}: e
xit status 1
stdout:


stderr:
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.50/containers/minikube/json": dial unix
/var/run/docker.sock: connect: permission denied

(base) admin1@bflcomp20:~/ai-kubernetes$ sudo usermod -aG docker $USER
[sudo] password for admin1:
(base) admin1@bflcomp20:~/ai-kubernetes$ docker ps
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.50/containers/json": dial unix /var/run/
docker.sock: connect: permission denied
(base) admin1@bflcomp20:~/ai-kubernetes$ newgrp docker
(base) admin1@bflcomp20:~/ai-kubernetes$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED         STATUS         PORTS
                                                                                    NAMES
b38cc9fa0b6c   gcr.io/k8s-minikube/kicbase:v0.0.50   "/usr/local/bin/entr…"   2 minutes ago   Up 2 minutes   127.0.0.1:32768->22/tcp, 127.0.0.1:32769->2376/tcp, 127.0.0.1:32770->5000/t
cp, 127.0.0.1:32771->8443/tcp, 127.0.0.1:32772->32443/tcp   minikube
(base) admin1@bflcomp20:~/ai-kubernetes$ minikube stop
minikube start
✋  Stopping node "minikube"  ...
🛑  Powering off "minikube" via SSH ...
🛑  1 node stopped.
😄  minikube v1.38.1 on Ubuntu 24.04
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🔄  Restarting existing docker container for "minikube" ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server
```

Mar 9 12:18

admin1@bflcomp20: ~/ai-kubernetes

admin1@bflcomp20: ~ | admin1@bflcomp20: ~/ai-kubernetes

```
Successfully installed blinker-1.9.0 click-8.3.1 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.6
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment inste
ad: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
 ---> Removed intermediate container 47ce4b43881b
 ---> 6856addb97be
Step 5/6 : EXPOSE 5000
 ---> Running in 0707bff4716c
 ---> Removed intermediate container 0707bff4716c
 ---> 11b9873b4d3b
Step 6/6 : CMD ["python","app.py"]
 ---> Running in fdeb6e444ee9
 ---> Removed intermediate container fdeb6e444ee9
 ---> b16ce3b0e5ef
Successfully built b16ce3b0e5ef
Successfully tagged ai-model:latest
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
error: the path "deployment.yaml" does not exist
error: the path "service.yaml" does not exist
(base) admin1@bflcomp20:~/ai-kubernetes$ nano deployment.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ nano service.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ ls
app.py  deployment.yaml  Dockerfile  service.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f deployment.yaml
deployment.apps/ai-model-deployment created
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f service.yaml
service/ai-model-service created
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl get pods
NAME                                 READY   STATUS    RESTARTS   AGE
ai-model-deployment-596f7f544-pbd5x   1/1     Running   0          12s
ai-model-deployment-596f7f544-ss2tm   1/1     Running   0          12s
(base) admin1@bflcomp20:~/ai-kubernetes$ minikube service ai-model-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	ai-model-service	5000	http://192.168.49.2:30007

```
🎉  Opening service default/ai-model-service in default browser...
(base) admin1@bflcomp20:~/ai-kubernetes$ Opening in existing browser session.
(base) admin1@bflcomp20:~/ai-kubernetes$
```

Mar 9 12:17

You are signed in as Dm | Inbox (2) - prathmesh.wa | Minikube Kubernetes Se | Kubernetes Dashboard | 192.168.49.2:30007

127.0.0.1:42083/api/v1/namespaces/kubernetes-dashboard/services/https:kubernetes-dashboard:/proxy/#/deployment?namespace=default

Gmail YouTube Maps

kubernetes | default | Search

Workloads > Deployments

Workloads
- Cron Jobs
- Daemon Sets
- Deployments
- Jobs
- Pods
- Replica Sets
- Replication Controllers
- Stateful Sets

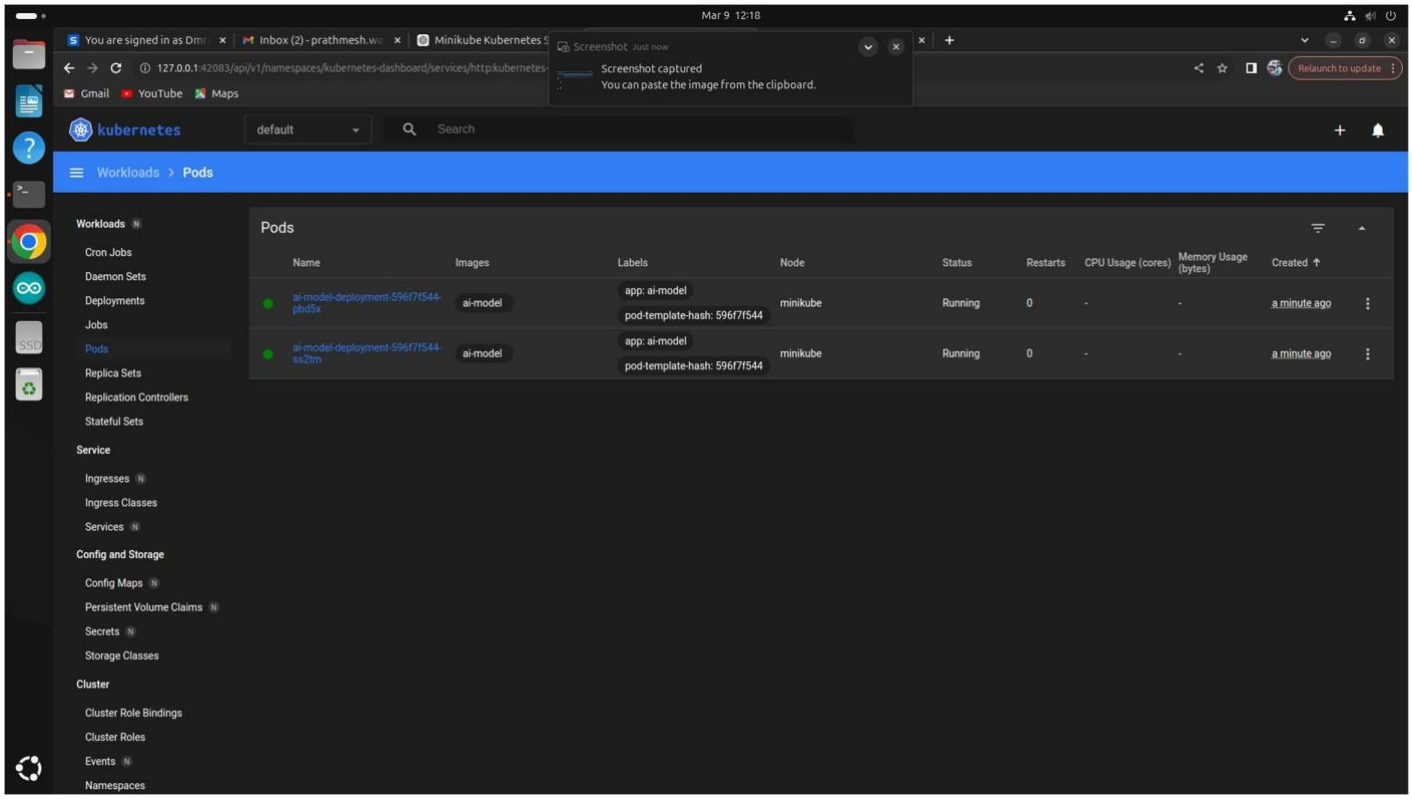
Service
- Ingresses
- Ingress Classes
- Services

Config and Storage
- Config Maps
- Persistent Volume Claims
- Secrets
- Storage Classes

Cluster
- Cluster Role Bindings
- Cluster Roles
- Events
- Namespaces

Deployments

Name	Images	Labels	Pods	Created
ai-model-deployment	ai-model	-	2 / 2	a minute ago


Mar 9 12:18
You are signed in as Dmr
Inbox (2) - prathmesh.wa
Minikube Kubernetes S
Screenshot Just now
Screenshot captured
You can paste the image from the clipboard.
127.0.0.1:42083/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-
Relaunch to update
Gmail
YouTube
Maps
kubernetes
default
Search
Workloads > Pods
Workloads
Cron Jobs
Daemon Sets
Deployments
Jobs
Pods
Replica Sets
Replication Controllers
Stateful Sets
Service
Ingresses
Ingress Classes
Services
Config and Storage
Config Maps
Persistent Volume Claims
Secrets
Storage Classes
Cluster
Cluster Role Bindings
Cluster Roles
Events
Namespaces
Pods
Name
Images
Labels
Node
Status
Restarts
CPU Usage (cores)
Memory Usage (bytes)
Created
ai-model-deployment-596f7f544-pbd5x
ai-model
app: ai-model
pod-template-hash: 596f7f544
minikube
Running
0
-
-
a minute ago
ai-model-deployment-596f7f544-ss2tm
ai-model
app: ai-model
pod-template-hash: 596f7f544
minikube
Running
0
-
-
a minute ago

Mar 9 12:23

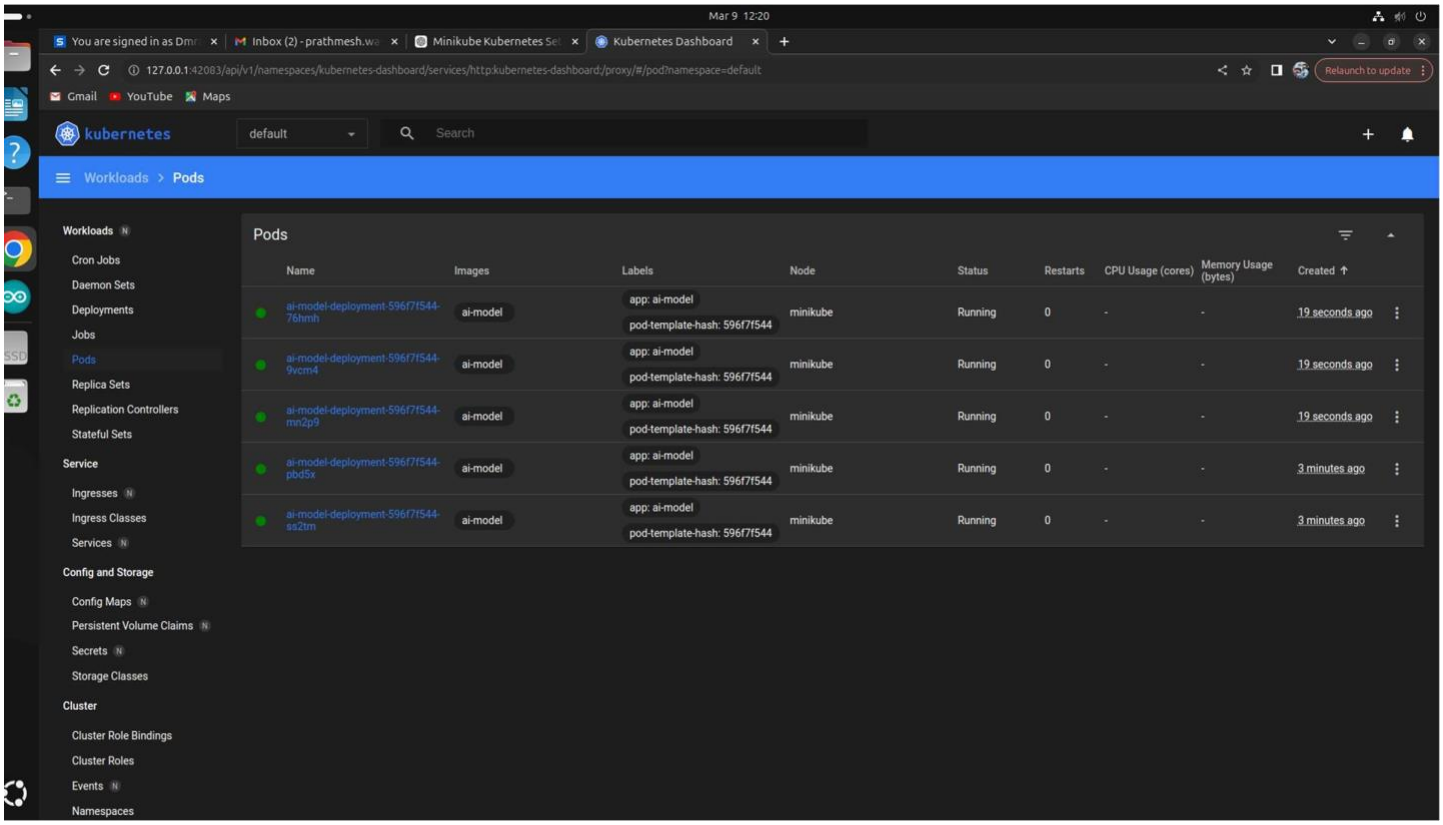
admin1@bflcomp20: ~/ai-kubernetes

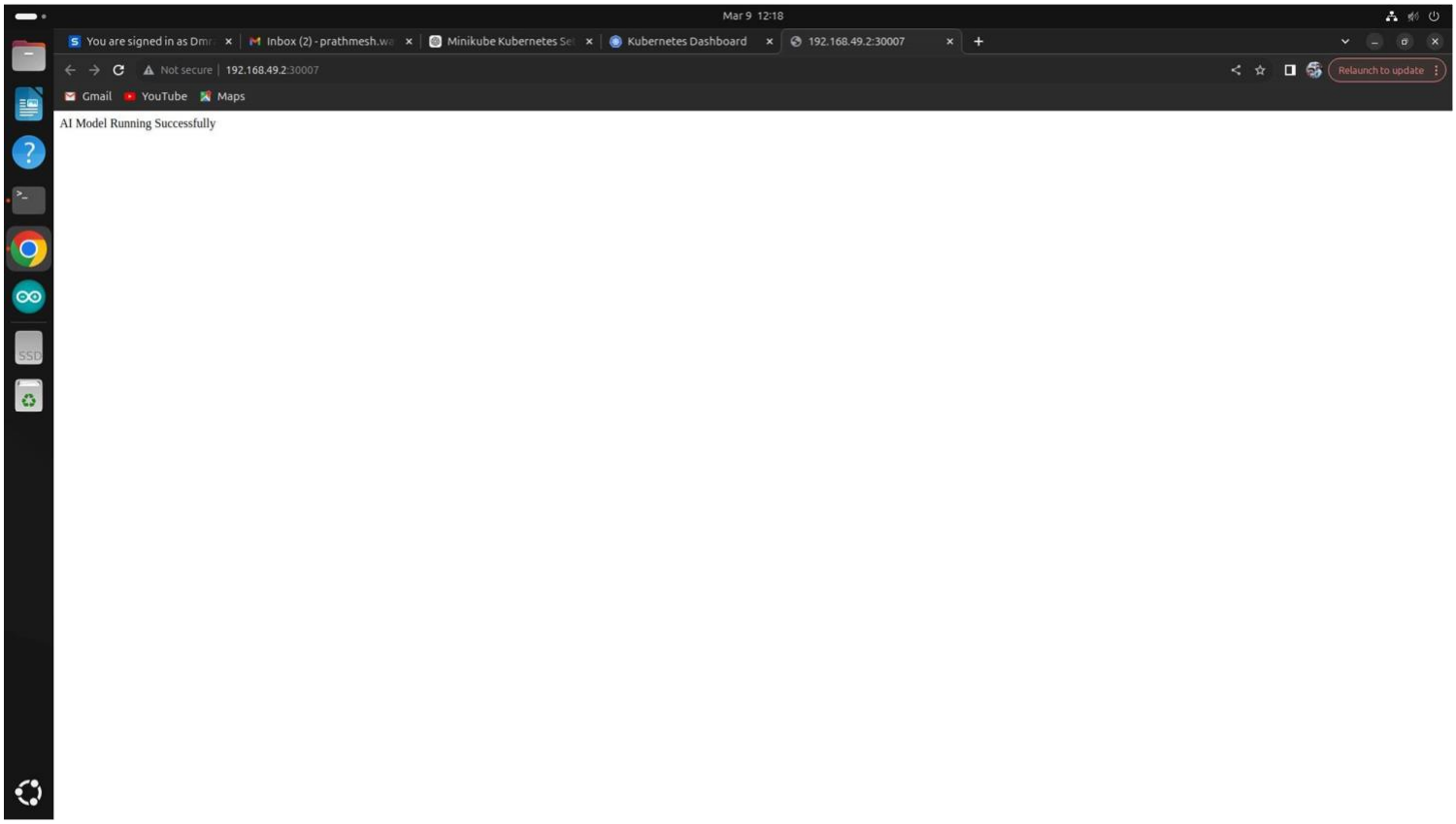
admin1@bflcomp20: ~ | admin1@bflcomp20: ~/ai-kubernetes

```
 ---> Running in fdeb6e444ee9
 ---> Removed intermediate container fdeb6e444ee9
 ---> b16ce3b0e5ef
Successfully built b16ce3b0e5ef
Successfully tagged ai-model:latest
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
error: the path "deployment.yaml" does not exist
error: the path "service.yaml" does not exist
(base) admin1@bflcomp20:~/ai-kubernetes$ nano deployment.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ nano service.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ ls
app.py  deployment.yaml  Dockerfile  service.yaml
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f deployment.yaml
deployment.apps/ai-model-deployment created
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl apply -f service.yaml
service/ai-model-service created
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
ai-model-deployment-596f7f544-pbd5x   1/1     Running   0          12s
ai-model-deployment-596f7f544-ss2tm   1/1     Running   0          12s
(base) admin1@bflcomp20:~/ai-kubernetes$ minikube service ai-model-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	ai-model-service	5000	http://192.168.49.2:30007

```
🎉  Opening service default/ai-model-service in default browser...
(base) admin1@bflcomp20:~/ai-kubernetes$ Opening in existing browser session.
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
ai-model-deployment-596f7f544-pbd5x   1/1     Running   0          3m28s
ai-model-deployment-596f7f544-ss2tm   1/1     Running   0          3m28s
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl scale deployment ai-model-deployment --replicas=5
deployment.apps/ai-model-deployment scaled
(base) admin1@bflcomp20:~/ai-kubernetes$ kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
ai-model-deployment-596f7f544-76hmh   1/1     Running   0          6s
ai-model-deployment-596f7f544-9vcm4   1/1     Running   0          6s
ai-model-deployment-596f7f544-mn2p9   1/1     Running   0          6s
ai-model-deployment-596f7f544-pbd5x   1/1     Running   0          3m46s
ai-model-deployment-596f7f544-ss2tm   1/1     Running   0          3m46s
(base) admin1@bflcomp20:~/ai-kubernetes$
```


Mar 9 12:20
You are signed in as Dmr
Inbox (2) - prathmesh.wa
Minikube Kubernetes Set
Kubernetes Dashboard
127.0.0.1:42083/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/#/pod?namespace=default
Relaunch to update
Gmail
YouTube
Maps
kubernetes
default
Search
Workloads > Pods
Workloads
Cron Jobs
Daemon Sets
Deployments
Jobs
Pods
Replica Sets
Replication Controllers
Stateful Sets
Service
Ingresses
Ingress Classes
Services
Config and Storage
Config Maps
Persistent Volume Claims
Secrets
Storage Classes
Cluster
Cluster Role Bindings
Cluster Roles
Events
Namespaces
Pods
Name
Images
Labels
Node
Status
Restarts
CPU Usage (cores)
Memory Usage (bytes)
Created
ai-model-deployment-596f7f544-76hmh
ai-model-deployment-596f7f544-9vcm4
ai-model-deployment-596f7f544-mn2p9
ai-model-deployment-596f7f544-pbd5x
ai-model-deployment-596f7f544-ss2tm
ai-model
app: ai-model
pod-template-hash: 596f7f544
minikube
Running
0
19 seconds ago
3 minutes ago


Mar 9 12:18
You are signed in as Dmr
Inbox (2) - prathmesh.wa
Minikube Kubernetes Set
Kubernetes Dashboard
192.168.49.2:30007
Not secure | 192.168.49.2:30007
Relaunch to update
Gmail
YouTube
Maps
AI Model Running Successfully